

JKBOSE 2026

CLASS 11TH

BIOLOGICAL CLASSIFICATION

LECTURE-04

BOTANY

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Topics To Covered

- 1** Kingdom Monera (Contd..) ✓

REPRODUCTION IN BACTERIA

Two types

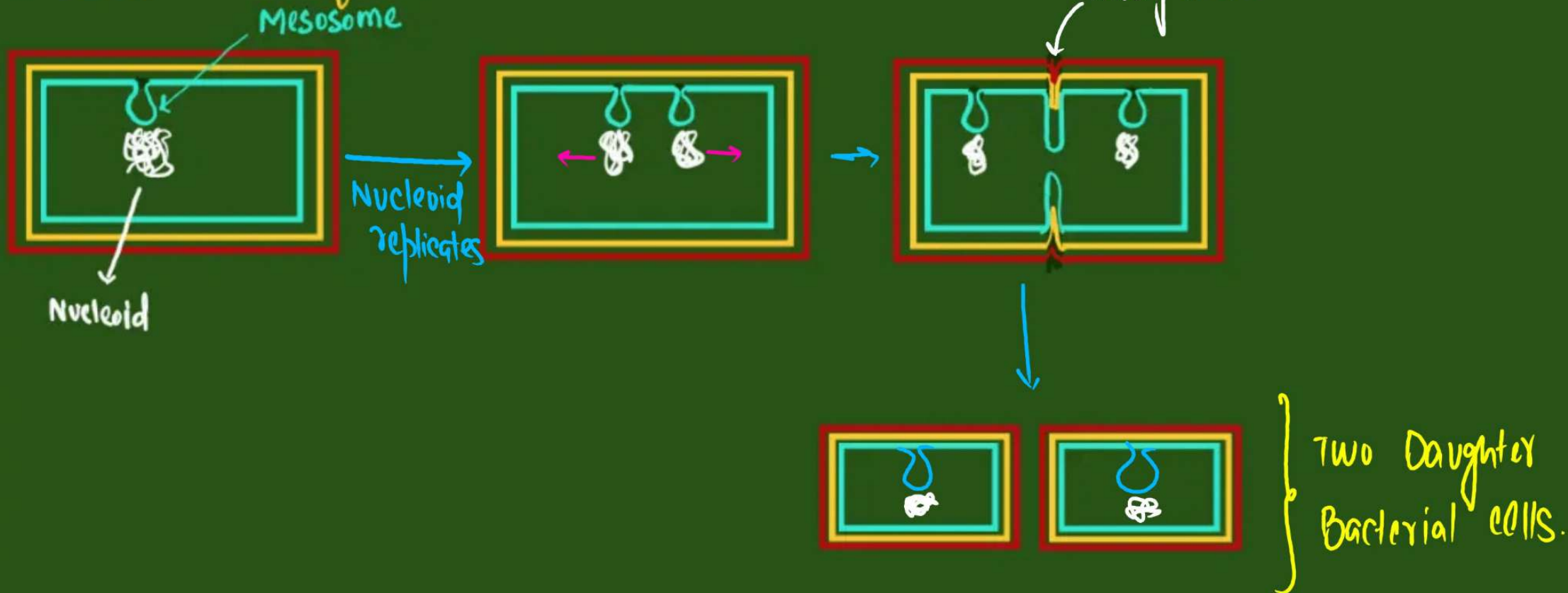
a. Asexual reproduction:-

b. Sexual reproduction:-

i. Binary fission:

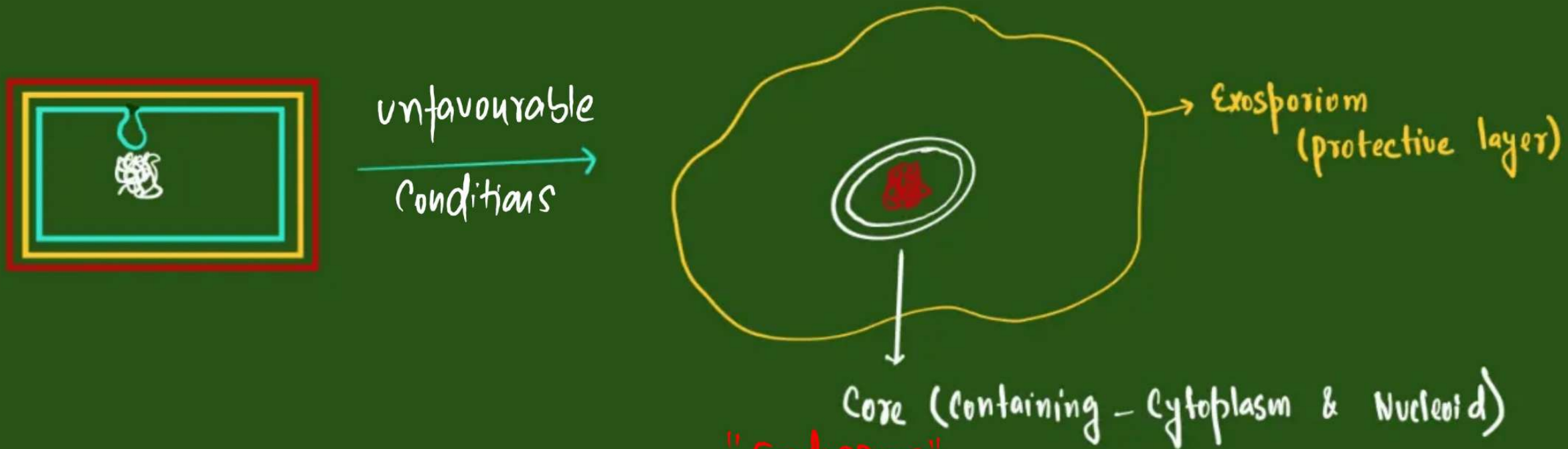
ii. Endospore formation.

- most common. ✓
- occurs under favourable conditions. ✓



ii. Endospore formation:-

- Occurs under Unfavourable Conditions.
- Endospore — Easily tolerates $\pm 100^{\circ}\text{C}$ temp., toxic chemicals, high pH etc.



≡ Under favourable Conditions — Exosporium gets degenerated.

≡ Endospore — liberated by air currents and upon germination forms a new bacterium.

ex. Bacillus, Clostridium ...

b. Sexual reproduction: (Genetic recombination)

- True Sexual reproduction absent.

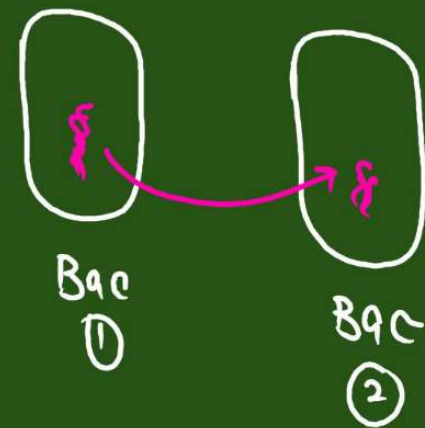
Reason:

No Sex organs
No gametogenesis
No fertilisation. (No Zygote formation)

≡ Sexual reproduction in Bacteria involves → variations by adopting a primitive type of DNA transfer from one bacterium to other. (Genetic Recombination)...

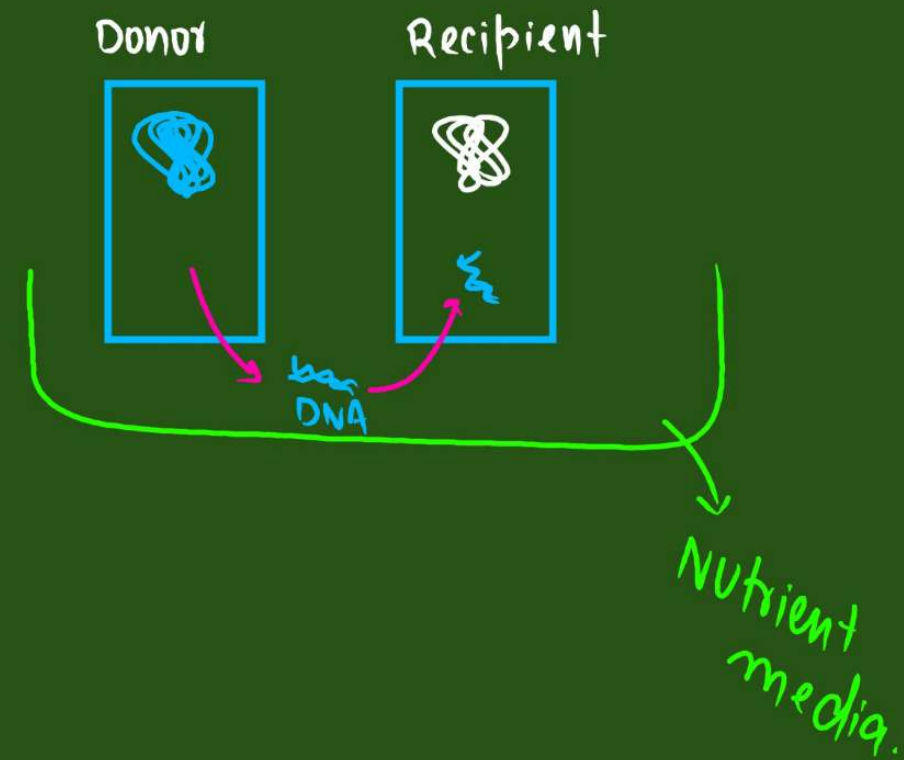
* DNA Recombination — takes place by 03 methods.

1. Transformation.
2. Transduction.
3. Conjugation.



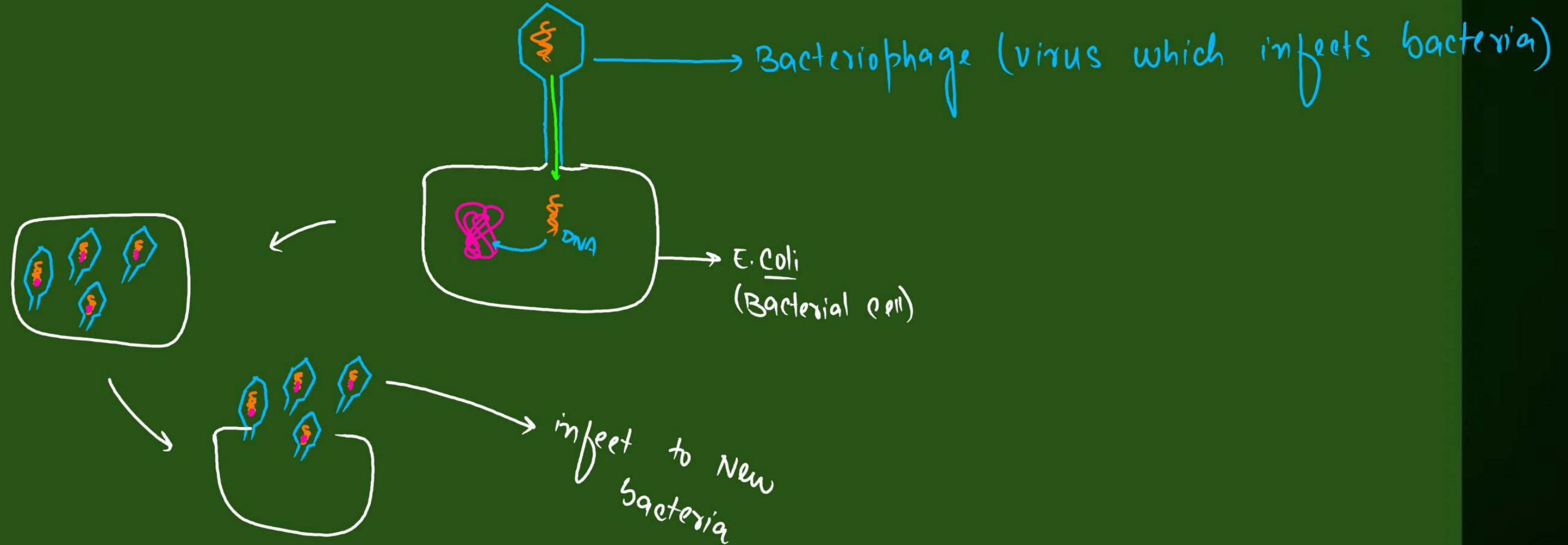
1. Transformation :-

- ≡ donor and recipient cell do not come in contact.
- ≡ Donor cell — releases piece of DNA and it is actively taken up by recipient cell from medium.



2. Transduction :

≡ Here DNA transfer from one bacterium to another takes place by means of Virus — Bacteriophage (virus which infects bacteria)

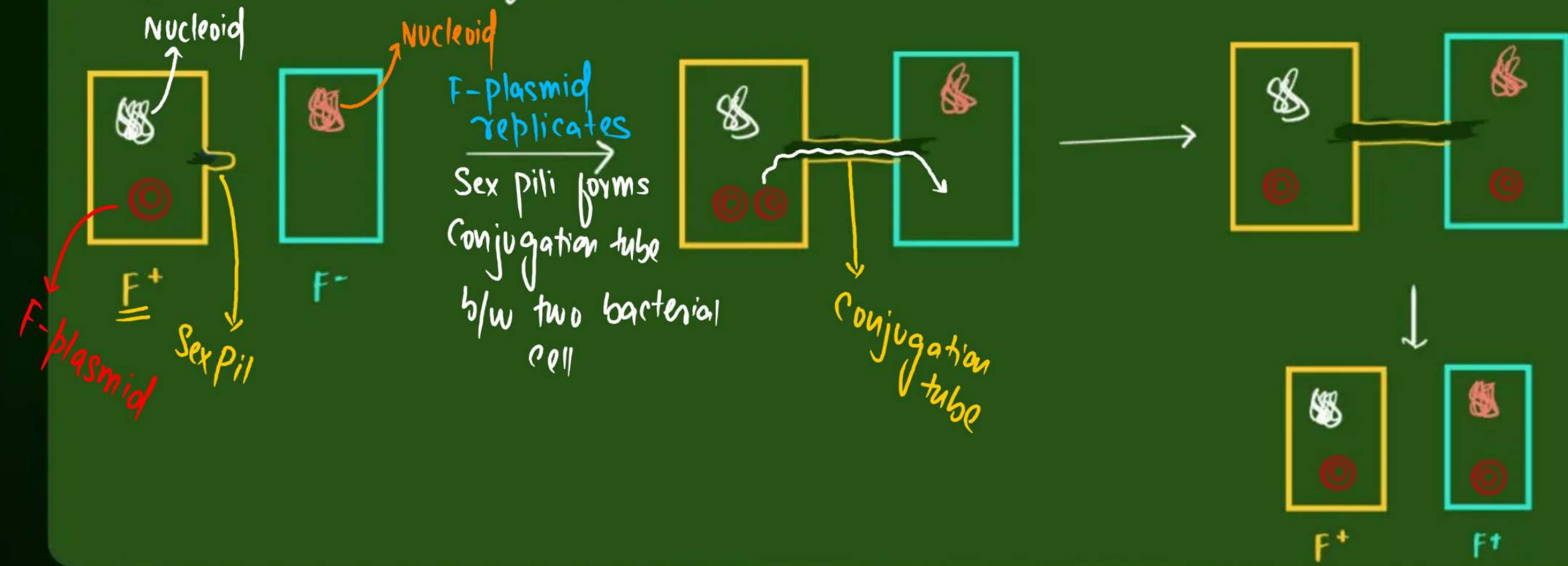


3. Conjugation :

≡ Two bacterial cells are involved

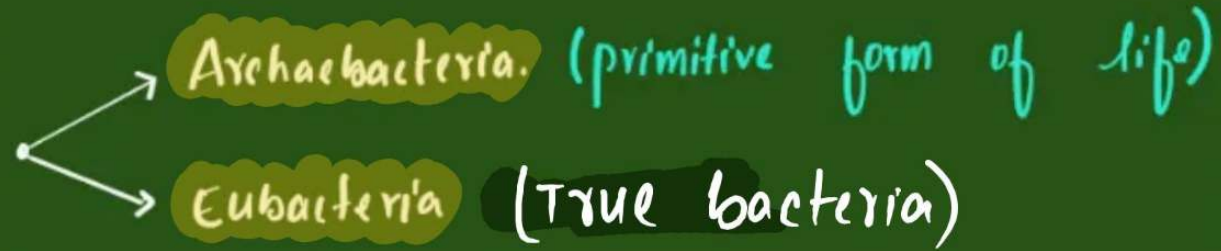
- F^+ (male) — Contains F-plasmid
 — acts as donor cell.
- F^- (female) — does not contain F-plasmid.
 — acts as recipient cell.

≡ F-plasmid contains "fertility factor" or "F-gene" — due to which sex pili are formed.



CLASSIFICATION OF MONERA

CLASSIFICATION OF MONERA:



* Archaeobacteria:

≡ Extreme environmental conditions (Habitat).

≡ Anaerobes — Survives in -nse of O_2 .

≡ Cell wall — Composed of pseudomurien.

≡ 03 types:

a. Methanogens: (obligate anaerobes)

≡ found in — Marshy habitats, ✓
 gut of ruminants, Sewage
 treatment plants. Cow,
 buffalo

≡ Methanogens — responsible for production of biogas. ✓

Ex. Methanobacterium.

b. Halophiles: (facultative anaerobes).

≡ found in Salt lakes (↑↑ Salt areas).

Ex. Halo bacterium

c. Thermoacidophiles: (facultative anaerobes)

≡ found in — Hot water Springs ($80-100^\circ C$).
 — low pH = 2 (acidic)

Ex. Thermus aquaticus.

* Eubacteria:

≡ True bacteria

Ex. Cyanobacteria, Mycoplasma

CYANOBACTERIA :- (Blue green Algae)

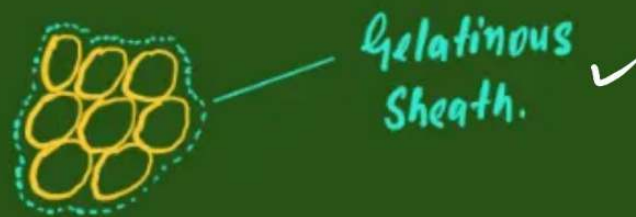
≡ Habitat — Fresh H₂O, Marine H₂O, Terrestrial.

≡ Body form

Unicellular

Filamentous

Colonial

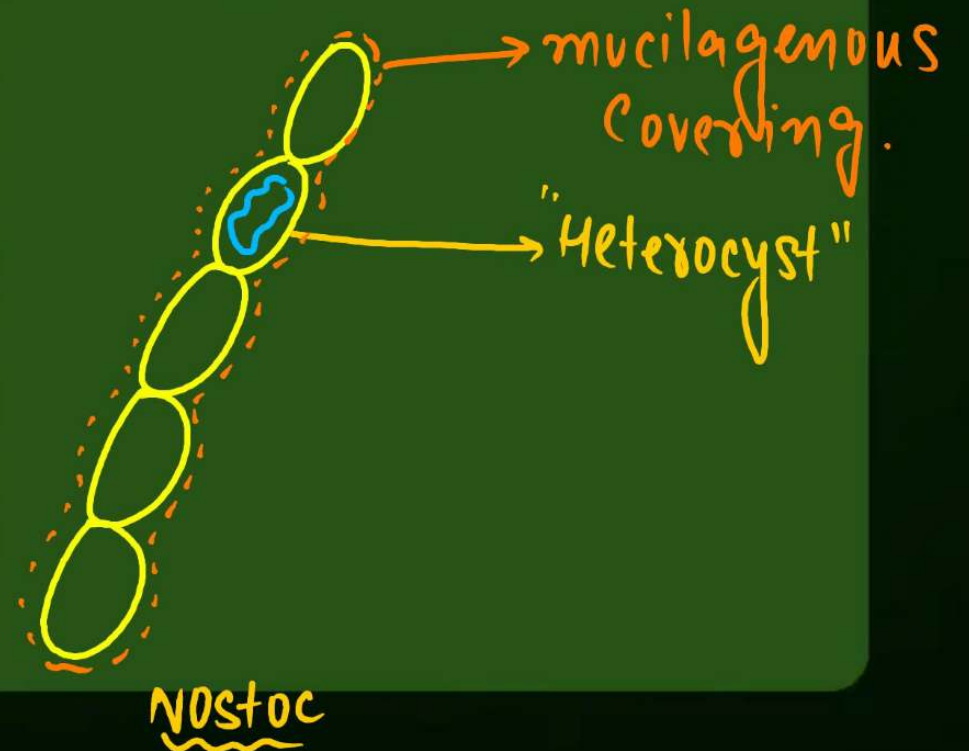


≡ Cell wall — composed of Peptidoglycan

≡ Chromatophores — +nt ≤ bears pigments (chl a etc)

≡ Nutrition — photosynthetic autotrophs.

* Some cyanobacteria fix atmospheric N₂ in specialised cells kld as "Heterocyst"
Ex. Nostoc, Anabaena.



WATER BLOOM :

H/w ??

Mycoplasma :-

- ≡ Smallest living cell (0.1 - 0.15 μ m)
- ≡ Discovered in pleural fluid of cattle suffering from pleuropneumonia disease.
- ≡ AKA MLO's (Mycoplasma like organisms).
- ≡ PPLO's (pleuropneumonia like organisms).
- ≡ Cell wall — Absent. (i.e why no fixed shape — pleiomorphic). → Continuously changes shape...
 - AKA "Jokers of plant kingdom"
 - Resistant to antibiotics (pencillin) — as it acts on cell wall
- ≡ Many are pathogenic in plants and animals.
 - ↓
disease causing

QUESTIONS

Which of the following is incorrect about Cyanobacteria? [MR* 2020-Covid]

Eubacteria

- ✓ 1 They lack heterocysts ✕
- 2 They often form blooms in polluted water bodies
- 3 They have chlorophyll 'a' similar to green plants
- 4 They are photoautotrophs

QUESTIONS

4/w

Which of the following are found in extreme saline conditions? [2017-Delhi]

- 1 Archaeobacteria
- 2 Eubacteria
- 3 Cyanobacteria
- 4 Mycobacteria

QUESTIONS

H/w



Methanogens belong to:

[2016-II]

- 1 Dinoflagellates
- 2 Slime moulds
- 3 Eubacteria
- 4 Archaeobacteria

QUESTIONS

H/W



The gut of cow and buffalo possess:

[2015]

- 1 Methanogens
- 2 *Cyanobacteria*
- 3 *Fucus*
- 4 *Chlorella*

QUESTIONS

H/W



Cell wall is absent in:

[2015 Re]

- 1 *Funaria*
- 2 *Mycoplasma*
- 3 *Nostoc*
- 4 *Aspergillus*

QUESTIONS

H/w



Archaeobacteria differ from Eubacteria in:

[2014]

- 1 Mode of reproduction
- 2 Cell membrane structure
- 3 Mode of nutrition
- 4 Cell shape

QUESTIONS

4/13



Which is wrong about Mycoplasma?

[2007]

- 1 They are called PPLO
- 2 They are pleomorphic
- 3 They are sensitive to penicillin
- 4 They cause diseases in plants

Thank
You